

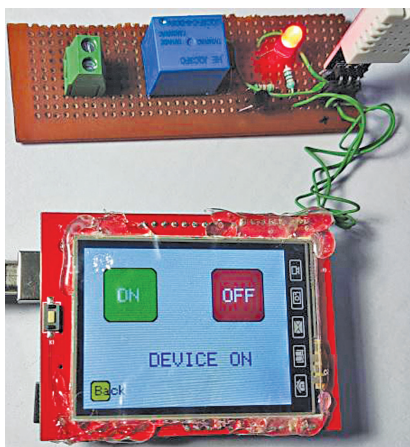


Fig. 8: Device on

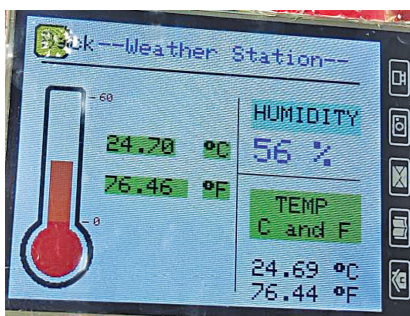


Fig. 9: Temperature and humidity readings

TouchScreen.h

One of the challenges in the project is calibrating the touch points in the touch-screen display.

Depending on the model and size of the touch-screen display, you need to calibrate x and y coordinates. The calibrated values in the code used by the author are shown in Fig. 6.

After uploading the code to Arduino Uno board, mount the TFT touch-screen on top of the Arduino board through the header pins already provided on Arduino board. You can carefully place it with a slight push on the touch-screen from top. Ensure the correct positions of the male header pins from touch-screen display are inserted into the corresponding female pins on the Arduino board.

Now you can connect the board to USB port of your PC/laptop or use a 9V-12V DC supply.

As soon as the power is on, the touch-screen display will turn on

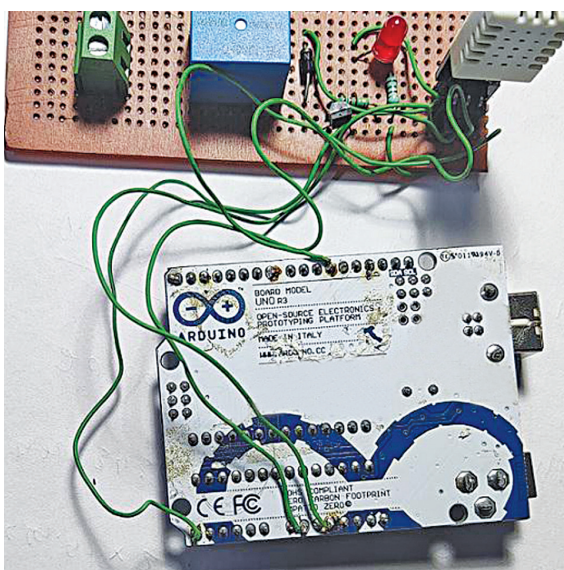


Fig. 10: Soldered connections

and you can see the main console panel on the TFT touch-screen, as shown in Fig. 7.

To control the device, such as light or air-conditioner connected across the relay, touch the 'Device Control' option. There are on and off buttons in Device Control section, as shown in Fig. 8. When you touch 'on' button, the LED will glow and at the same time the electrical device/appliance connected across the relay will turn on. Touch the 'off' button to turn it off.

If you touch the 'Weather Station' option, it will take you to another screen, as shown in Fig. 9. Here, you can see the current humidity and temperature values captured by the DHT22 sensor. The humidity here is basically the relative humidity expressed in percentage, which represents the amount of water vapour in the air at a given temperature compared with the maximum possible water vapour at that same temperature. Here the

temperature values are read in both °C and °F units. So, no conversion is required.

Construction and testing

The circuit can be assembled on a breadboard or Veroboard. The author used a 10x3cm size Veroboard to mount the relay, sensor, transistor, LED, 2-pin terminal connector (optional), and the two resistors.

Since most of the Arduino pins are used by the TFT touch-screen

display, you need external wires to solder at digital pin 11 of Arduino for the relay driver and analogue pin A5 for the DHT22, as shown in Fig. 10. You also need to solder 5V and GND pins of Arduino for power supply to the Veroboard. If you are using DHT22 sensor instead of the module, make sure you connect a 5-kilo-ohm pull-up resistor at its output pin, otherwise you may not get correct temperature and humidity readings.

Also note that some 6cm TFT touch-screen displays may not work. So, it is recommended to use the model shown in Fig. 3. If you want to control an appliance like cooler or AC, you need to replace the sugar-cube relay with a high-power relay having contact ratings above 16A.

Though you can power the circuit by connecting it to USB port of your PC, you can also power it with external power supply through the DC jack on the Arduino board. The author used a 9V DC power adaptor to power the Arduino board. When power is connected to the circuit you will get the main console screen, as shown in Fig. 7. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com

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